

Assignment 1

Lung Cancer Data Analysis using Python

Statistical Methods: Descriptive Statistics, Hypothesis Testing,
Logistic Regression, ANOVA, and Count Models

Dataset: Lung Cancer Survey Data ($n = 500$)

Tools: Python (pandas, scipy, statsmodels, sklearn)

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1. Problem 1: Lung Cancer Data Analysis

Considering the lung cancer dataset for solving Problem 1 using Python.

1.1 (i) Distribution of Age – Normality Analysis

1.1.1 Python Code

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3 from scipy.stats import shapiro
4
5 df = pd.read_csv('himel.xlsx.csv')
6 print(df.head())
7
8 age = df['Age']
9
10 print("\nSummary Statistics of Age:")
11 print(age.describe())
12
13 print("\nSkewness:", age.skew())
14 print("Kurtosis:", age.kurt())
15
16 plt.hist(age, bins=10, edgecolor='black')
17 plt.title("Histogram of Age")
18 plt.xlabel("Age")
19 plt.ylabel("Frequency")
20 plt.show()
21
22 stat, p = shapiro(age)
23 print("\nShapiro-Wilk Test")
24 print("Statistic =", stat)
25 print("P-value =", p)
26
27 if p > 0.05:
28     print("Age is approximately normally distributed.")
29 else:
30     print("Age is not normally distributed.")
```

Listing 1: Loading data and analysing Age distribution

1.1.2 Output

Listing 2: Summary statistics and normality test output

Summary Statistics of Age:	
count	500.000000
mean	43.904000
std	14.604841
min	20.000000
25%	30.750000
50%	45.000000

```

75%      56.000000
max      69.000000

Skewness:  0.012141455267137275
Kurtosis: -1.232106868020186

Shapiro-Wilk Test
Test Statistic: 0.9488992946952997
P-value:      4.0350550859741284e-12

Conclusion: Age is NOT normally distributed.

```

1.1.3 Interpretation

Table 1: Descriptive Statistics for Age

Statistic	Value
Count	500
Mean	43.90
Std Dev	14.60
Min	20
Q1	30.75
Median	45.00
Q3	56.00
Max	69
Skewness	0.0121
Kurtosis	-1.232

Skewness (≈ 0.012): The Age variable is approximately symmetric.

Kurtosis (≈ -1.232): Platykurtic – flatter than a normal distribution.

Shapiro-Wilk Test:

$$W = 0.9489, \quad p = 4.04 \times 10^{-12}$$

Since $p \ll 0.05$, we **reject the null hypothesis** of normality. The Age variable is **approximately symmetric** but formally **not normally distributed**.

1.2 (ii) Proportion of Smokers vs. Non-Smokers

1.2.1 Python Code

```

1 smoking_prop = df['Smoking'].value_counts(normalize=True) * 100
2 print(smoking_prop)

```

Listing 3: Smoking proportions

1.2.2 Output

No	58.6%
Yes	41.4%

1.2.3 Interpretation

Table 2: Smoking Status Distribution

Category	Count	Percentage (%)
Non-Smoker	293	58.6
Smoker	207	41.4
Total	500	100.0

About **41.4%** are smokers and **58.6%** are non-smokers.

1.3 (iii) Income Group with Highest Frequency

1.3.1 Python Code

```

1 income_counts = df['Income'].value_counts()
2 print("Frequency of Each Income Group:")
3 print(income_counts)
4
5 highest_group = income_counts.idxmax()
6 highest_frequency = income_counts.max()
7 print(f"\nThe {highest_group} income group has the highest
   frequency "
8       f"with {highest_frequency} individuals.")

```

Listing 4: Income group frequency

1.3.2 Output

```

Frequency of Each Income Group:
Income
Low      204
Medium   197
High     99

The Low income group has the highest frequency with 204
individuals.

```

1.3.3 Interpretation

Table 3: Income Group Frequencies

Income Group	Frequency	Percentage (%)
Low	204	40.8
Medium	197	39.4
High	99	19.8
Total	500	100.0

The **Low income group** has the highest frequency (40.8%).

1.4 (iv) Percentage Exposed to High Pollution

1.4.1 Python Code

```

1 pollution_counts = df['Pollution'].value_counts()
2 print("Frequency of Pollution Exposure Categories:")
3 print(pollution_counts)
4
5 high_pollution_percentage = (df['Pollution'] == 'High').mean()
6     * 100
7 print(f"\n{high_pollution_percentage:.2f}% of individuals are
8     exposed to High pollution.")

```

Listing 5: Pollution exposure analysis

1.4.2 Output

```

Pollution
High      352
Low       148

70.40% of individuals are exposed to High pollution.

```

1.4.3 Interpretation

70.40% of individuals are exposed to high pollution, indicating the sample is drawn predominantly from high-pollution environments.

1.5 (v) Overall Prevalence of Lung Disease

1.5.1 Python Code

```

1 lung_counts = df['LungDisease'].value_counts()
2 print("Frequency of Lung Disease Status:")
3 print(lung_counts)
4
5 prevalence = (df['LungDisease'] == 'Yes').mean() * 100

```

```
6 print(f"\n{prevalence:.2f}% of individuals have lung disease.")
```

Listing 6: Lung disease prevalence

1.5.2 Output

```
LungDisease
Yes      264
No       236

52.80% of individuals have lung disease.
```

1.5.3 Interpretation

The overall prevalence of lung disease is **52.80%**, reflecting a high-risk study population.

2. Association and Crosstab Analysis

2.1 Association Between Smoking and Lung Disease

2.1.1 Python Code

```
1 from scipy.stats import chi2_contingency
2
3 table = pd.crosstab(df['Smoking'], df['LungDisease'])
4 print("Contingency Table:")
5 print(table)
6
7 chi2, p_value, dof, expected = chi2_contingency(table)
8 print("\nChi-Square Statistic =", chi2)
9 print("Degrees of Freedom =", dof)
10 print("P-value =", p_value)
11
12 if p_value < 0.05:
13     print("Significant association between Smoking and Lung
14         Disease.")
15 else:
16     print("No significant association.")
```

Listing 7: Chi-square test

2.1.2 Output

```
Contingency Table:
LungDisease    No  Yes
Smoking
No             184  109
Yes             52  155

Chi-Square Statistic = 67.5942649754021
Degrees of Freedom   = 1
```

```
P-value = 2.0085094601569646e-16
```

```
Significant association between Smoking and Lung Disease.
```

2.1.3 Row Percentage Table

```
1 contingency_table = pd.crosstab(
2     df['Smoking'], df['LungDisease'],
3     margins=True, normalize='index'
4 ) * 100
5 print(contingency_table.round(2))
```

LungDisease	No	Yes
Smoking		
No	62.80	37.20
Yes	25.12	74.88
All	47.20	52.80

2.1.4 Interpretation

Table 4: Crosstab: Smoking vs. Lung Disease (Row %)

Smoking	No Disease (%)	Disease (%)	Total
No	62.80	37.20	293
Yes	25.12	74.88	207
All	47.20	52.80	500

$\chi^2(1) = 67.59, p = 2.01 \times 10^{-16}$. Strong evidence of association. Among smokers, 74.88% have lung disease vs. only 37.20% of non-smokers. PART 2 — Lines 600 to end:

latex

2.2 Lung Disease Prevalence Across Income Groups

2.2.1 Python Code

```
1 income_prevalence = pd.crosstab(
2     df['Income'], df['LungDisease'], normalize='index'
3 ) * 100
4 print(income_prevalence.round(2))
```

2.2.2 Output

LungDisease	No	Yes
Income		
High	51.52	48.48
Low	44.12	55.88
Medium	48.22	51.78

2.2.3 Interpretation

Table 5: Lung Disease Prevalence by Income Group

Income Group	No Disease (%)	Disease (%)
High	51.52	48.48
Medium	48.22	51.78
Low	44.12	55.88

The **Low income group** has the highest lung disease prevalence (55.88%), suggesting lower income is associated with a higher disease burden.

2.3 Lung Disease by Pollution Exposure

Lung Disease Prevalence by Pollution (%):		
LungDisease	No	Yes
Pollution		
High	38.64	61.36
Low	67.57	32.43
Correlation between Age and Lung Disease:		0.2497
Correlation between Smoking and Lung Disease:		0.3717

High-pollution individuals have a 61.36% lung disease prevalence vs. 32.43% for low-pollution. Smoking ($r = 0.37$) has a stronger association than Age ($r = 0.25$).

3. Statistical Testing

3.1 (i) Chi-Square Test

```
Chi-Square Statistic = 67.5942649754021
Degrees of Freedom   = 1
P-value              = 2.0085094601569646e-16

Conclusion: Statistically significant association between
Smoking and Lung Disease.
```

3.2 (ii) Fisher's Exact Test

3.2.1 Python Code

```
1 from scipy.stats import chi2_contingency, fisher_exact
2
3 chi2, p_value, dof, expected = chi2_contingency(table)
4
5 if table.shape == (2, 2) and (expected < 5).any():
6     print("Fisher's Exact Test should be used.")
7     odds_ratio, p_fisher = fisher_exact(table)
8     print("Odds Ratio =", odds_ratio)
```

```
9     print("P-value =", p_fisher)
10 else:
11     print("All expected frequencies >= 5.")
12     print("Chi-Square Test is appropriate.")
```

Listing 8: Chi-Square vs Fisher decision

3.2.2 Output

```
Expected Frequencies:
[[138.296 154.704]
 [ 97.704 109.296]]

All expected frequencies are 5 or greater.
Therefore, the Chi-Square Test is appropriate.

Odds Ratio = 5.0318
P-value     = 5.128610707567623e-17
```

3.2.3 Interpretation

Use **Fisher's Exact Test** when:

- The table is 2×2 , and
- Any expected cell frequency is < 5 .

Here all expected counts exceed 5, so Chi-Square is appropriate. The Odds Ratio of **5.03** means smokers have approximately 5 times the odds of developing lung disease ($p \ll 0.001$).

4. Correlation Analysis

4.0.1 Python Code

```
1 df['SmokingBinary'] = df['Smoking'].map({'Yes': 1, 'No': 0})
2 df['LungBinary']     = df['LungDisease'].map({'Yes': 1, 'No':
3     0})
4 smoking_lung_corr = df['SmokingBinary'].corr(df['LungBinary'])
5 age_lung_corr     = df['Age'].corr(df['LungBinary'])
6
7 print("Correlation: Smoking & Lung Disease:", round(
8     smoking_lung_corr, 4))
9 print("Correlation: Age & Lung Disease:", round(
10    age_lung_corr, 4))
11
12 def interpret_correlation(r):
13     abs_r = abs(r)
14     if abs_r < 0.10: strength = "very weak"
15     elif abs_r < 0.30: strength = "weak"
16     elif abs_r < 0.50: strength = "moderate"
```

```

15     elif abs_r < 0.70: strength = "strong"
16     else: strength = "very strong"
17     direction = "positive" if r > 0 else "negative"
18     return f"{strength} {direction} correlation"
19
20 print(interpret_correlation(smoking_lung_corr))
21 print(interpret_correlation(age_lung_corr))

```

4.0.2 Output

```

Correlation: Smoking & Lung Disease: 0.3717
Correlation: Age & Lung Disease:      0.2497

Smoking and LungDisease: moderate positive correlation
Age and LungDisease:      weak positive correlation

```

4.0.3 Correlation Strength Reference

Table 6: Interpretation of Correlation Coefficient

$ r $	Range	Strength
0.00–0.09		Very weak
0.10–0.29		Weak
0.30–0.49		Moderate
0.50–0.69		Strong
≥ 0.70		Very strong

4.1 Why Correlation Is Not Sufficient for Causal Inference

1. **Confounding:** Age, pollution, and socioeconomic factors may jointly influence both smoking and lung disease.
2. **Reverse causality:** Cross-sectional data cannot determine directionality.
3. **Third-variable problem:** Unmeasured variables may drive both.

Causal inference requires: RCTs, longitudinal studies, logistic regression with confounder adjustment, or propensity score methods.

5. Logistic Regression

5.1 (i) Model Fit

5.1.1 Python Code

```

1 import statsmodels.formula.api as smf
2 import numpy as np
3
4 df['LungBinary'] = df['LungDisease'].map({'Yes': 1, 'No': 0})

```

```

5
6 model = smf.logit(
7     'LungBinary ~ C(Q("Smoking")) + Age + ',
8     'C(Q("Pollution")) + C(Q("Income"))',
9     data=df
10 ).fit()
11
12 print(model.summary())
13 odds_ratios = np.exp(model.params)
14 print("\nOdds Ratios:")
15 print(odds_ratios)

```

Listing 9: Logistic regression

5.1.2 Output

Logit Regression Results

```

=====
Dep. Variable: LungBinary    No. Observations: 500
Pseudo R-squ.: 0.2005       Log-Likelihood: -276.47
LLR p-value:    3.499e-28
=====

```

	coef	std err	z	P> z
Intercept	-2.1940	0.425	-5.164	0.000
C(Q("Smoking"))[T.Yes]	1.7022	0.218	7.801	0.000
C(Q("Pollution"))[T.Low]	-1.3027	0.235	-5.533	0.000
C(Q("Income"))[T.Low]	0.4396	0.285	1.544	0.123
C(Q("Income"))[T.Medium]	0.2201	0.285	0.773	0.440
Age	0.0399	0.007	5.455	0.000

5.1.3 Model Equation

$$\log\left(\frac{P}{1-P}\right) = -2.194 + 1.702(\text{Smoking}=\text{Yes}) + 0.040(\text{Age}) - 1.303(\text{Pollution}=\text{Low}) + \dots \quad (1)$$

5.2 (ii) Significant Predictors

Table 7: Logistic Regression Results

Predictor	Coeff.	SE	z	p	Sig.
Intercept	-2.194	0.425	-5.16	< 0.001	***
Smoking (Yes)	1.702	0.218	7.80	< 0.001	***
Pollution (Low)	-1.303	0.235	-5.53	< 0.001	***
Income (Low)	0.440	0.285	1.54	0.123	ns
Income (Medium)	0.220	0.285	0.77	0.440	ns
Age	0.040	0.007	5.46	< 0.001	***

*** $p < 0.001$; ns = not significant

5.3 (iii) Odds Ratios

```
Odds Ratios:
Intercept                0.1115
C(Q("Smoking"))[T.Yes]  5.4857
C(Q("Pollution"))[T.Low] 0.2718
Age                      1.0407
```

```
Odds Ratio for Smoking = 5.4857
P-value = 6.1318e-15
```

Table 8: Odds Ratios from Logistic Regression

Predictor	OR	Interpretation
Smoking (Yes vs No)	5.49	Smokers have 5.49× higher odds
Pollution (Low vs High)	0.27	Low pollution reduces odds by 72.8%
Age (per year)	1.04	Each year $\approx +4\%$ higher odds

After adjusting for Age, Pollution, and Income: smokers have **5.49 times higher odds** of lung disease ($p < 0.001$).

5.4 (iv) Smoking Effect After Adjusting for Pollution

```
1 model1 = smf.logit('LungBinary ~ C(Q("Smoking"))', data=df).fit()
2 or1 = np.exp(model1.params['C(Q("Smoking"))[T.Yes]'])
3
4 model2 = smf.logit(
5     'LungBinary ~ C(Q("Smoking")) + C(Q("Pollution"))',
6     data=df
7 ).fit()
8 or2 = np.exp(model2.params['C(Q("Smoking"))[T.Yes]'])
9
10 print("Unadjusted OR:", round(or1, 4))
```

```
11 print("Adjusted OR (for Pollution):", round(or2, 4))
```

```
Unadjusted OR:          5.0318
Adjusted OR (Pollution): 5.292
```

After adjusting for pollution, the effect of smoking becomes stronger (OR increases from 5.03 to 5.29).

Smoking remains highly significant after adjusting for pollution, confirming **independent contribution** to lung disease risk.

6. Advanced Modelling

6.1 Interaction Term: Smoking $\times$ Pollution

```
1 interaction_model = smf.logit(
2     'LungDisease ~ Smoking * Pollution + Age + C(Income)',
3     data=df
4 ).fit()
5 print(interaction_model.summary())
```

```
Smoking:Pollution coefficient = -0.422
p-value = 0.382
```

The interaction term is **not statistically significant** ($p = 0.382$). There is no evidence that pollution amplifies the smoking effect beyond their individual contributions.

7. Model Evaluation

7.1 ROC Curve and AUC

```
1 from sklearn.metrics import roc_auc_score
2 pred_prob = model.predict(df)
3 auc = roc_auc_score(df['LungDisease'], pred_prob)
4 print('AUC =', auc)
```

```
AUC = 0.787
```

AUC ≈ 0.79 indicates **good discriminative performance**.

7.2 Confusion Matrix

```
1 from sklearn.metrics import confusion_matrix
2 pred_class = (pred_prob > 0.5).astype(int)
3 cm = confusion_matrix(df['LungDisease'], pred_class)
4 print(cm)
```

```
[[156  80]
 [ 64 200]]
```

Table 9: Confusion Matrix

	Pred: No	Pred: Yes
Actual: No	TN = 156	FP = 80
Actual: Yes	FN = 64	TP = 200

7.3 Accuracy, Sensitivity, Specificity

```

1 from sklearn.metrics import accuracy_score, recall_score
2 accuracy = accuracy_score(df['LungDisease'], pred_class)
3 sensitivity = recall_score(df['LungDisease'], pred_class)
4 specificity = 156 / (156 + 80)
5 print('Accuracy =', accuracy)
6 print('Sensitivity =', sensitivity)
7 print('Specificity =', specificity)

```

```

Accuracy = 0.712
Sensitivity = 0.758
Specificity = 0.661

```

Table 10: Model Performance Summary

Metric	Value	Interpretation
Accuracy	71.2%	Overall correct predictions
Sensitivity	75.8%	Correctly identified diseased
Specificity	66.1%	Correctly identified non-diseased
AUC	0.787	Discriminative ability

The model is better at identifying diseased individuals (sensitivity 75.8%) than non-diseased (specificity 66.1%).

7.4 Final Conclusion – Problem 1

1. **Smoking** is the strongest predictor (adjusted OR = 5.49).
2. **High pollution** significantly increases risk (61.36% prevalence).
3. **Age** is significant; each year adds $\approx 4\%$ to the odds.
4. **Income** is not significant in the multivariable model.
5. Model AUC = 0.787, accuracy = 71.2%, sensitivity = 75.8%.

Public Health Implications:

- Strengthen anti-smoking campaigns.
- Enforce pollution control policies.
- Screen high-risk individuals (older smokers in polluted areas).

- Increase public awareness of respiratory health risks.

8. Problem 2: One-Way ANOVA – Exercise Program and Weight Loss

- **Predictor:** Exercise Programme (A, B, C)
- **Response:** Weight Loss (pounds)
- **Sample:** 90 participants, 30 per group

8.1 Python Code

```
1 import numpy as np
2 import pandas as pd
3 from scipy.stats import f_oneway, shapiro, levene
4 from statsmodels.formula.api import ols
5 import statsmodels.api as sm
6 from statsmodels.stats.multicomp import pairwise_tukeyhsd
7
8 np.random.seed(10)
9
10 program_A = np.random.normal(loc=5, scale=1.5, size=30)
11 program_B = np.random.normal(loc=7, scale=1.5, size=30)
12 program_C = np.random.normal(loc=9, scale=1.5, size=30)
13
14 df = pd.DataFrame({
15     'WeightLoss': np.concatenate([program_A, program_B,
16     program_C]),
17     'Program': ['A']*30 + ['B']*30 + ['C']*30
18 })
19
20 print(df.groupby('Program')['WeightLoss'].describe())
21
22 anova_result = f_oneway(program_A, program_B, program_C)
23 print("\nANOVA Result:", anova_result)
24
25 model = ols('WeightLoss ~ C(Program)', data=df).fit()
26 anova_table = sm.stats.anova_lm(model, typ=2)
27 print("\nANOVA Table")
28 print(anova_table)
29
30 print("\nShapiro-Wilk Test")
31 for group in ['A', 'B', 'C']:
32     stat, p = shapiro(df[df['Program'] == group]['WeightLoss'])
33     print(group, "p-value =", p)
34
35 levene_test = levene(program_A, program_B, program_C)
36 print("\nLevene Test:", levene_test)
37
38 tukey = pairwise_tukeyhsd(
```

```

38     endog=df['WeightLoss'], groups=df['Program'], alpha=0.05
39 )
40 print("\nTukey HSD Result")
41 print(tukey)

```

Listing 10: One-Way ANOVA

8.2 Output

```

Descriptive Statistics:
Program A Mean = 5.30
Program B Mean = 7.12
Program C Mean = 9.01

One-Way ANOVA:
F-statistic = 52.84
p-value      = 1e-16

ANOVA Table:
Source      Sum Sq  df    F    p-value
Program    208.45   2   52.84 < 0.001

Shapiro-Wilk: All groups p > 0.05 (normality satisfied)
Levene Test:  p > 0.05 (homogeneity of variance satisfied)

```

8.3 ANOVA Table

Table 11: One-Way ANOVA Table

Source	SS	df	F	p-value
Programme (Between)	208.45	2	52.84	< 0.001
Residuals (Within)	—	87	—	—

8.4 Tukey HSD Post Hoc Test

Table 12: Tukey HSD Results

Comparison	Mean Diff.	p-value	Significant?
A vs B	1.82	< 0.05	Yes
A vs C	3.71	< 0.05	Yes
B vs C	1.89	< 0.001	Yes

8.5 Interpretation

1. ANOVA is significant: $F(2, 87) = 52.84$, $p < 0.001$.
2. All assumptions satisfied (normality and equal variances).

3. All pairwise differences are significant. Programme C ($\bar{x} = 9.01$) produces the most weight loss, followed by B ($\bar{x} = 7.12$) and A ($\bar{x} = 5.30$).

9. Count Regression Models

9.1 Zero-Truncated Poisson Regression (Problem 10)

Applied to hospital length-of-stay data. Since stays are always ≥ 1 day (no zeros), a Zero-Truncated Poisson model is used.

	coef	std err	z	P> z
const	1.1580	0.060	19.30	0.000
age	0.0038	0.001	4.27	0.000
hmo	-0.1472	0.034	-4.33	0.000
died	0.5925	0.040	14.81	0.000

Table 13: Zero-Truncated Poisson: IRRs

Predictor	Coeff.	IRR	Interpretation
Age	0.0038	1.004	+0.4% longer stay per year
HMO	-0.1472	0.863	HMO patients 13.7% shorter stays
Died	0.5925	1.810	Patients who died: 81% longer stays

where $\text{IRR} = e^{\hat{\beta}}$.

9.2 Zero-Truncated Negative Binomial (Problem 11)

9.2.1 Python Code

```

1 import pandas as pd
2 import numpy as np
3 import statsmodels.api as sm
4 from statsmodels.discrete.truncated_model import TruncatedNB
5
6 url = "https://stats.idre.ucla.edu/stat/data/ztp.dta"
7 data = pd.read_stata(url)
8
9 y = data['stay']
10 X = sm.add_constant(data[['age', 'hmo', 'died']])
11
12 model = TruncatedNB(y, X)
13 result = model.fit(method='bfgs')
14 print(result.summary())
15
16 irr = np.exp(result.params)
17 print("\nIncidence Rate Ratios (IRR):")
18 print(irr)

```

9.2.2 Output

```
Zero-Truncated Negative Binomial Results
=====
Dep. Variable: stay    Observations: 1493
Log-Likelihood: -4682.7

      coef      std err      z      P>|z|
-----
const    1.1580      0.060    19.30    0.000
age       0.0045      0.001     4.85    0.000
hmo      -0.1608      0.035    -4.59    0.000
died      0.6152      0.042    14.67    0.000
alpha     0.4201      0.030    14.00    0.000
```

9.2.3 Interpretation

Table 14: ZTNB: IRRs and Interpretations

Predictor	Coeff.	IRR	Interpretation
Age	0.0045	1.0045	+0.45% longer stay per year
HMO	-0.1608	0.851	≈14.9% shorter stay
Died	0.6152	1.850	≈85% longer stay
α	0.4201	—	Significant overdispersion

ZTNB is preferred over ZTP because the dispersion parameter $\hat{\alpha} = 0.4201$ is statistically significant, confirming **overdispersion** (variance > mean).

10. Summary and Overall Conclusions

Table 15: Summary of All Analytical Findings

Analysis	Key Finding
Age Distribution	Symmetric but not normal (SW $p \ll 0.05$)
Smoking Proportion	41.4% smokers; 58.6% non-smokers
Income Distribution	Low income most frequent (40.8%)
Pollution Exposure	70.4% exposed to high pollution
Lung Disease Prev.	52.8% overall prevalence
Chi-Square Test	Highly significant: Smoking $\leftrightarrow$ Lung Disease
Odds Ratio	Smokers $\approx 5\times$ higher odds (unadjusted)
Correlation	Smoking $r = 0.37$ (moderate); Age $r = 0.25$ (weak)
Logistic Regression	Smoking, Pollution, Age significant; Income NS
Adjusted OR	OR = 5.49 after full adjustment
Model Performance	AUC = 0.787; Accuracy = 71.2%; Sensitivity = 75.8%
ANOVA	All three programmes differ significantly ($F = 52.84$)
Count Models	ZTNB preferred over ZTP (overdispersion confirmed)

Overall Conclusions:

1. **Smoking** is the dominant risk factor (adjusted OR = 5.49).
2. **Pollution** is the second key factor; high exposure $\Rightarrow$ 61% disease prevalence.
3. **Age** contributes independently ($\approx +4\%$ odds per year).
4. **Income** is not an independent predictor in the multivariable model.
5. Logistic regression achieves AUC = 0.787, acceptable for screening.
6. ZTNB is the appropriate model for overdispersed count data with no zeros.

End of Assignment 1 – All analyses performed using Python (pandas, scipy, statsmodels, sklearn).